

SATURN: Splitting Approach on Tensorized Ubiquitous Regression Networks

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Paper ID 12215

Abstract. *Fully Connected* (FC) layers are often used in many neural network architectures, such as *Convolutional Neural Networks* (CNNs), which usually flatten the data before feeding it to the FC layers. Despite empirical success, some drawbacks include the changing of topological structure and the number of required parameters. Methods like the *Tensor Contraction Layer* (TCL) and *Tensor Regression Layer* (TRL) are capable candidates to use instead of FC layers, which preserve the multilinear structure of data as well as reduce the cost of execution. To exploit their capability even further, our method uses specific splitting approaches to generate thinner, higher dimensional tensors from the original data to reduce the cost by a greater factor while also maintaining accuracy. Our experiments on CIFAR10 and MNIST, applied to known models such as ResNet, VGG, and other typical CNNs, demonstrate our method's ability to reduce the cost compared to previous models and their TCL-TRL adaptation while also maintaining accuracy.

Keywords: Machine Learning, Tensorized Regression Layer, Tensorized Contraction Layer, Patching and Splitting, Tensor Methods, Low-Rank Approximation

1 Introduction

Conventional Neural Networks, especially those that are applied to images such as *Convolutional Neural Network* (CNNs), often consist of two main parts: One that is responsible to extract features from the input data, and the other part is a *MultiLayer Perceptron* (MLP) which Maps the extracted features to the desired space. Despite the importance of data and their spatial relationship, maintaining their topological structure is often discarded since many known CNNs like ResNet or VGG, seem to work fine as they are; However in cases where these structures are more important than the others such as MRI, It is crucial to adapt the known networks to work without destroying the multi-linear structure of the data.

Multi-linear or Multi-modal structures have always existed in many natural datasets and we can represent these using Tensors; Images could be considered as a 3rd orders tensors with width, height and the color channels as the corresponding 3 modes, the same could be considered for videos as 4th order tensor with an extra time mode. In practice, many of the natural datasets can be considered as tensors which then requires the tensor expansion to the linear algebra.

Considering the new advances in *Deep Neural Networks* (DNNs) and their applications to computer vision such as image recognition, image segmentation, image generation, video generation, vision transformers, etc ...; most architectures still flatten the data before using them on MLPs which not only destroys the natural structure of the data but also increase the number of parameters needed for the model.

Images as the prime example of data used in vision machine learning tasks are considered 3rd order tensors which require their own algebra to be applied when fed to the MLPs (convolution layers are already designed to work on tensors). *Fully Connected* (FC) layers used in many architectures are the main culprit behind vectorizing the data and the number of parameters. To alleviate the situation, a better replacement for FC layers is needed which not only preserves the current model's accuracy but also reduces the number of required parameters and does not destroy the structure of data.

Some works have been done to address these issues which will be discussed in Section 2. Our method provides a better solution based on one of the previous works and the experiments on CIFAR10 and MNIST applied to known architectures, ResNet and VGG, and typical CNNs proves our superiority. Using a splitting method, to split the original tensor along its modes into smaller tensors and stacking them into a newer, thinner and higher dimensional tensor we reduced the number of parameters while also maintaining the accuracy.

2 Related Works

2.1 Language

All manuscripts must be in English.

2.2 Paper length

Papers submitted for review should be complete. The length should match that intended for final publication. Papers accepted for the conference will be allocated 14 pages (plus references) in the proceedings. Note that the allocated 14 pages do not include the references. The reason for this policy is that we do not want authors to omit references for sake of space limitations.

Papers with more than 14 pages (excluding references) will be rejected without review. This includes papers where the margins and formatting are deemed to have been significantly altered from those laid down by this style guide. The reason such papers will not be reviewed is that there is no provision for supervised revisions of manuscripts. The reviewing process cannot determine the suitability of the paper for presentation in 14 pages if it is reviewed in 16.

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All lines should be numbered, as in this example document. This makes reviewing more efficient, because reviewers can refer to a line on a page. If you are preparing a document using a non-L^AT_EX document preparation system, please arrange for an equivalent line numbering.

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Please number all of your sections and displayed equations. Again, this makes reviewing more efficient, because reviewers can refer to a line on a page. Also, it is important for readers to be able to refer to any particular equation. Just because you didn't refer to it in the text doesn't mean some future reader might not need to refer to it. It is cumbersome to have to use circumlocutions like "the equation second from the top of page 3 column 1". (Note that the line numbering will not be present in the final copy, so is not an alternative to equation numbers). Some authors might benefit from reading Mermin's description of how to write mathematics: www.pamitc.org/documents/mermin.pdf .

3 Blind review

Many authors misunderstand the concept of anonymizing for blind review. Blind review does not mean that one must remove citations to one's own work. In fact it is often impossible to review a paper unless the previous citations are known and available.

Blind review means that you do not use the words "my" or "our" when citing previous work. That is all. (But see below for technical reports).

Saying "this builds on the work of Lucy Smith [1]" does not say that you are Lucy Smith, it says that you are building on her work. If you are Smith and Jones, do not say "as we show in [7]", say "as Smith and Jones show in [7]" and at the end of the paper, include reference 7 as you would any other cited work.

An example of a bad paper:

An analysis of the frobnicatable foo filter.

In this paper we present a performance analysis of our previous paper [1], and show it to be inferior to all previously known methods. Why the previous paper was accepted without this analysis is beyond me.

[1] Removed for blind review

An example of an excellent paper:

An analysis of the frobnicatable foo filter.

In this paper we present a performance analysis of the paper of Smith [1], and show it to be inferior to all previously known methods. Why the previous paper was accepted without this analysis is beyond me.

[1] Smith, L. and Jones, C. "The frobnicatable foo filter, a fundamental contribution to human knowledge". Nature 381(12), 1-213.

If you are making a submission to another conference at the same time, which covers similar or overlapping material, you may need to refer to that submission in order to explain the differences, just as you would if you had previously published related work. In such cases, include the anonymized parallel submission [1] as additional material and cite it as

1. Authors. “The frobnicatable foo filter”, BMVC 2014 Submission ID 324, Supplied as additional material `bmvc14.pdf`.

Finally, you may feel you need to tell the reader that more details can be found elsewhere, and refer them to a technical report. For conference submissions, the paper must stand on its own, and not *require* the reviewer to go to a techreport for further details. Thus, you may say in the body of the paper “further details may be found in [2]”. Then submit the techreport as additional material. Again, you may not assume the reviewers will read this material.

Sometimes your paper is about a problem which you tested using a tool which is widely known to be restricted to a single institution. For example, let’s say it’s 1969, you have solved a key problem on the Apollo lander, and you believe that the ECCV audience would like to hear about your solution. The work is a development of your celebrated 1968 paper entitled “Zero-g frobnication: How being the only people in the world with access to the Apollo lander source code makes us a wow at parties”, by Zeus.

You can handle this paper like any other. Don’t write “We show how to improve our previous work [Anonymous, 1968]. This time we tested the algorithm on a lunar lander [name of lander removed for blind review]”. That would be silly, and would immediately identify the authors. Instead write the following:

We describe a system for zero-g frobnication. This system is new because it handles the following cases: A, B. Previous systems [Zeus et al. 1968] didn’t handle case B properly. Ours handles it by including a foo term in the bar integral.

...

The proposed system was integrated with the Apollo lunar lander, and went all the way to the moon, don’t you know. It displayed the following behaviours which show how well we solved cases A and B: ...

As you can see, the above text follows standard scientific convention, reads better than the first version, and does not explicitly name you as the authors. A reviewer might think it likely that the new paper was written by Zeus, but cannot make any decision based on that guess. He or she would have to be sure that no other authors could have been contracted to solve problem B.

For sake of anonymity, it’s recommended to omit acknowledgements in your review copy. They can be added later when you prepare the final copy.

4 Manuscript Preparation

This is an edited version of Springer LNCS instructions adapted for ECCV 2016 first paper submission. You are strongly encouraged to use $\text{\LaTeX}2_{\epsilon}$ for the preparation of your camera-ready manuscript together with the corresponding Springer class file `llnsc.cls`.

We would like to stress that the class/style files and the template should not be manipulated and that the guidelines regarding font sizes and format should be adhered to. This is to ensure that the end product is as homogeneous as possible.

4.1 Printing Area

The printing area is 122 mm $\times$ 193 mm. The text should be justified to occupy the full line width, so that the right margin is not ragged, with words hyphenated as appropriate. Please fill pages so that the length of the text is no less than 180 mm.

4.2 Layout, Typeface, Font Sizes, and Numbering

Use 10-point type for the name(s) of the author(s) and 9-point type for the address(es) and the abstract. For the main text, please use 10-point type and single-line spacing. We recommend using Computer Modern Roman (CM) fonts, Times, or one of the similar typefaces widely used in photo-typesetting. (In these typefaces the letters have serifs, i.e., short endstrokes at the head and the foot of letters.) Italic type may be used to emphasize words in running text. Bold type and underlining should be avoided. With these sizes, the interline distance should be set so that some 45 lines occur on a full-text page.

Headings. Headings should be capitalized (i.e., nouns, verbs, and all other words except articles, prepositions, and conjunctions should be set with an initial capital) and should, with the exception of the title, be aligned to the left. Words joined by a hyphen are subject to a special rule. If the first word can stand alone, the second word should be capitalized. The font sizes are given in Table 1.

Here are some examples of headings: “Criteria to Disprove Context-Freeness of Collage Languages”, “On Correcting the Intrusion of Tracing Non-deterministic Programs by Software”, “A User-Friendly and Extendable Data Distribution System”, “Multi-flip Networks: Parallelizing GenSAT”, “Self-determinations of Man”.

Lemmas, Propositions, and Theorems. The numbers accorded to lemmas, propositions, and theorems etc. should appear in consecutive order, starting with the number 1, and not, for example, with the number 11.

Table 1. Font sizes of headings. Table captions should always be positioned *above* the tables. The final sentence of a table caption should end without a full stop

Heading level	Example	Font size and style
Title (centered)	Lecture Notes ...	14 point, bold
1st-level heading	1 Introduction	12 point, bold
2nd-level heading	2.1 Printing Area	10 point, bold
3rd-level heading	Headings. Text follows ...	10 point, bold
4th-level heading	<i>Remark.</i> Text follows ...	10 point, italic

4.3 Figures and Photographs

Please produce your figures electronically and integrate them into your text file. For L^AT_EX users we recommend using package **graphicx** or the style files **psfig** or **epsf**.

Check that in line drawings, lines are not interrupted and have constant width. Grids and details within the figures must be clearly readable and may not be written one on top of the other. Line drawings should have a resolution of at least 800 dpi (preferably 1200 dpi). For digital halftones 300 dpi is usually sufficient. The lettering in figures should have a height of 2 mm (10-point type). Figures should be scaled up or down accordingly. Please do not use any absolute coordinates in figures.

Figures should be numbered and should have a caption which should always be positioned *under* the figures, in contrast to the caption belonging to a table, which should always appear *above* the table. Please center the captions between the margins and set them in 9-point type (Fig. 1 shows an example). The distance between text and figure should be about 8 mm, the distance between figure and caption about 5 mm.

If possible (e.g. if you use L^AT_EX) please define figures as floating objects. L^AT_EX users, please avoid using the location parameter “h” for “here”. If you have to insert a pagebreak before a figure, please ensure that the previous page is completely filled.

4.4 Formulas

Displayed equations or formulas are centered and set on a separate line (with an extra line or halfline space above and below). Displayed expressions should be numbered for reference. The numbers should be consecutive within the contribution, with numbers enclosed in parentheses and set on the right margin. For example,

$$\psi(u) = \int_0^T \left[\frac{1}{2} (A_0^{-1}u, u) + N^*(-u) \right] dt$$

(1)

$$= 0?$$

(2)

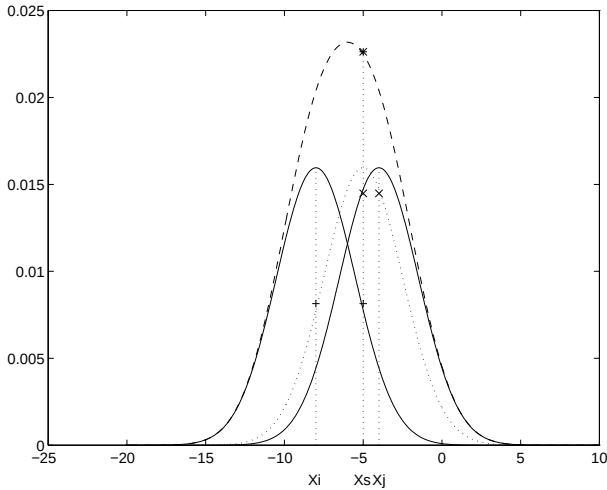


Fig. 1. One kernel at x_s (*dotted kernel*) or two kernels at x_i and x_j (*left and right*) lead to the same summed estimate at x_s . This shows a figure consisting of different types of lines. Elements of the figure described in the caption should be set in italics, in parentheses, as shown in this sample caption. The last sentence of a figure caption should generally end without a full stop

Please punctuate a displayed equation in the same way as ordinary text but with a small space before the end punctuation.

4.5 Footnotes

The superscript numeral used to refer to a footnote appears in the text either directly after the word to be discussed or, in relation to a phrase or a sentence, following the punctuation sign (comma, semicolon, or full stop). Footnotes should appear at the bottom of the normal text area, with a line of about 2 cm in $\text{\TeX}$ and about 5 cm in Word set immediately above them.¹

4.6 Program Code

Program listings or program commands in the text are normally set in typewriter font, e.g., CMTT10 or Courier.

Example of a Computer Program

```
program Inflation (Output)
```

```
{Assuming annual inflation rates of 7%, 8%, and 10%,...
```

¹ The footnote numeral is set flush left and the text follows with the usual word spacing. Second and subsequent lines are indented. Footnotes should end with a full stop.

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315 years};
316 const
317     MaxYears = 10;
318 var
319     Year: 0..MaxYears;
320     Factor1, Factor2, Factor3: Real;
321 begin
322     Year := 0;
323     Factor1 := 1.0; Factor2 := 1.0; Factor3 := 1.0;
324     WriteLn('Year  7% 8% 10%'); WriteLn;
325     repeat
326         Year := Year + 1;
327         Factor1 := Factor1 * 1.07;
328         Factor2 := Factor2 * 1.08;
329         Factor3 := Factor3 * 1.10;
330         WriteLn(Year:5,Factor1:7:3,Factor2:7:3,Factor3:7:3)
331     until Year = MaxYears
332 end.
```

333 (Example from Jensen K., Wirth N. (1991) Pascal user manual and report. Springer, 333
334 New York) 334
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337 **4.7 Citations** 337

338 The list of references is headed “References” and is not assigned a number in 338
339 the decimal system of headings. The list should be set in small print and placed 339
340 at the end of your contribution, in front of the appendix, if one exists. Please do 340
341 not insert a pagebreak before the list of references if the page is not completely 341
342 filled. An example is given at the end of this information sheet. For citations in 342
343 the text please use square brackets and consecutive numbers: [3], [4], [5] ... 343
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345 **5 Conclusions** 345
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347 The paper ends with a conclusion. 347
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Now we have reached the maximum size of the ECCV 2016 submission (excluding references). References should start immediately after the main text, but can continue on p.15 if needed.

References

1. Authors: The frobnicatable foo filter (2014) BMVC14 submission ID 324. Supplied as additional material **bmvc14.pdf**.
2. Authors: Frobnication tutorial (2014) Supplied as additional material **tr.pdf**.
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